

EE227 – Digital Logic Design

Lecture 22

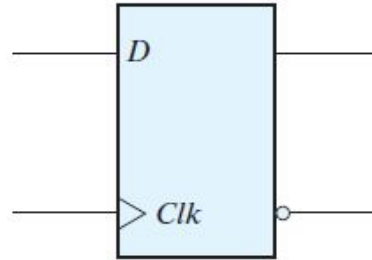
Outline

- Analysis of Sequential Circuits
 - State Equations
 - State Table
 - State Diagram
- Flip Flop Input Equations
 - Analysis with D Flip Flop
 - Analysis with JK Flip Flop
 - Analysis with T – Flip Flop

Summary of Flip Flops

- D Flip Flop

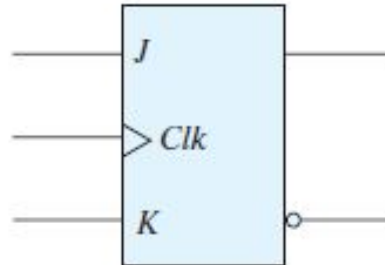
$$Q(t + 1) = D$$



D	$Q(t + 1)$
0	0 Reset
1	1 Set

- JK Flip Flop

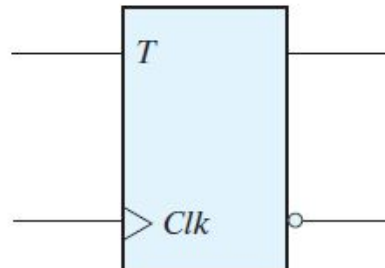
$$Q(t + 1) = JQ' + K'Q$$



J	K	$Q(t + 1)$
0	0	$Q(t)$ No change
0	1	0 Reset
1	0	1 Set
1	1	$Q'(t)$ Complement

- T Flip Flop

$$Q(t + 1) = T \oplus Q$$



T	$Q(t + 1)$
0	$Q(t)$ No change
1	$Q'(t)$ Complement

Analysis of Sequential Circuits (1/2)

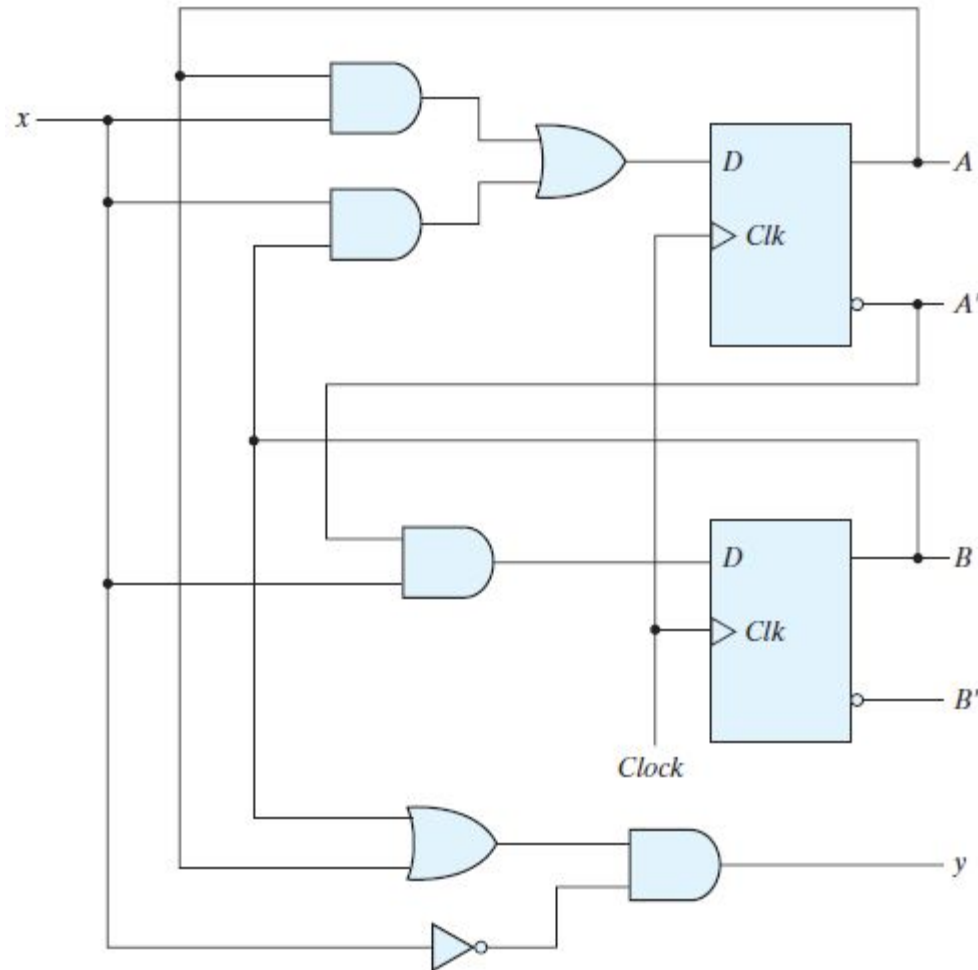
- Analysis describes what a given circuit will do under certain operating conditions
- The behavior of a clocked sequential circuit is determined from the inputs, the outputs, and the state of its flip-flops
- The analysis can be done with the help of
 - State equations
 - State tables
 - State diagrams
- The typical sequence of analysis is as follow
Circuit $\Rightarrow$ State Equations $\Rightarrow$ State Table $\Rightarrow$ State Diagram
- Where circuit and state equations are interchangeable

Analysis of Sequential Circuits (2/2)

- State Equations
 - A state equation (also called a transition equation) specifies the next state as a function of the present state and inputs
 - Can be obtained directly from the part of sequential circuit which is forming a combinational circuit
- State Table
 - Also called the transition table
 - The table consists of four sections labeled present state, input, next state, and output
 - The next states are determined from state equations
- State Diagram
 - State diagram shows all possible states and transfer from one state to another depending upon previous state and input combinations

Example 1

- Analyze the following sequential circuit



Example 1 (Solution)

- State Equations

$$A(t+1) = A(t)x(t) + B(t)x(t)$$

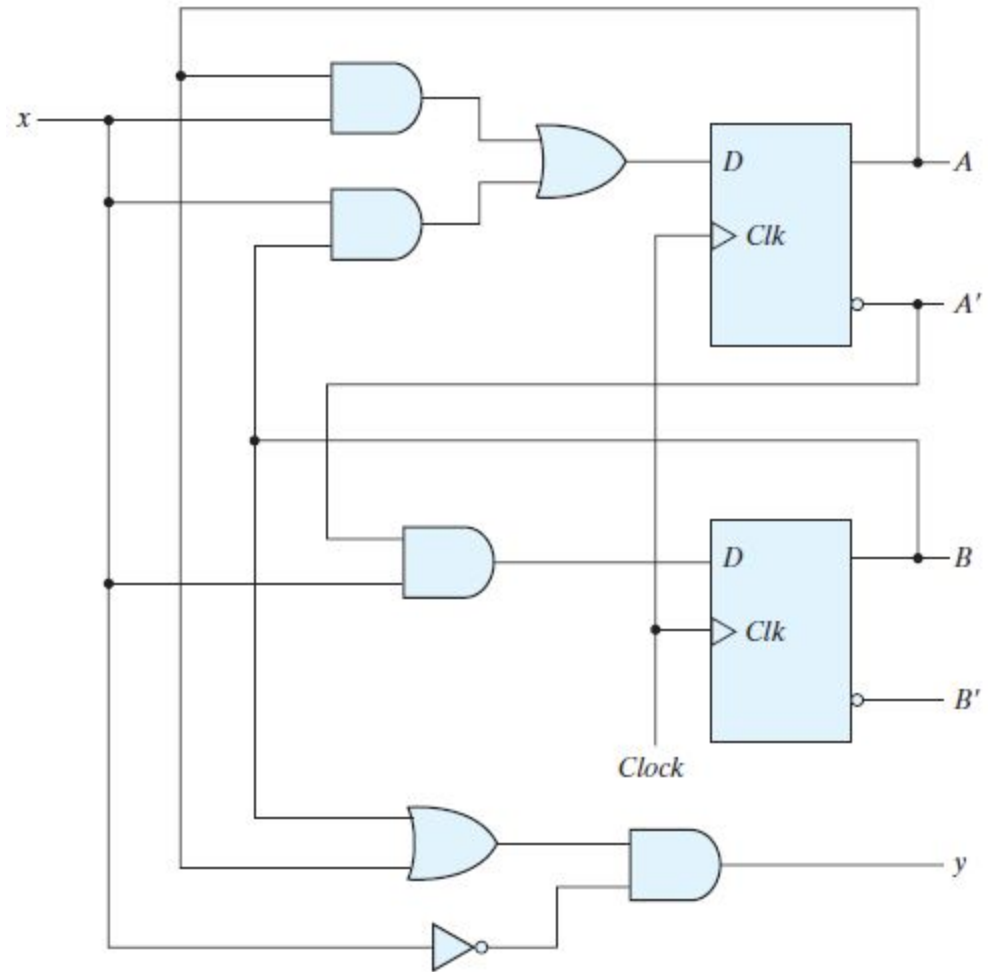
$$B(t+1) = A'(t)x(t)$$

$$A(t+1) = Ax + Bx$$

$$B(t+1) = A'x$$

$$y(t) = [A(t) + B(t)]x'(t)$$

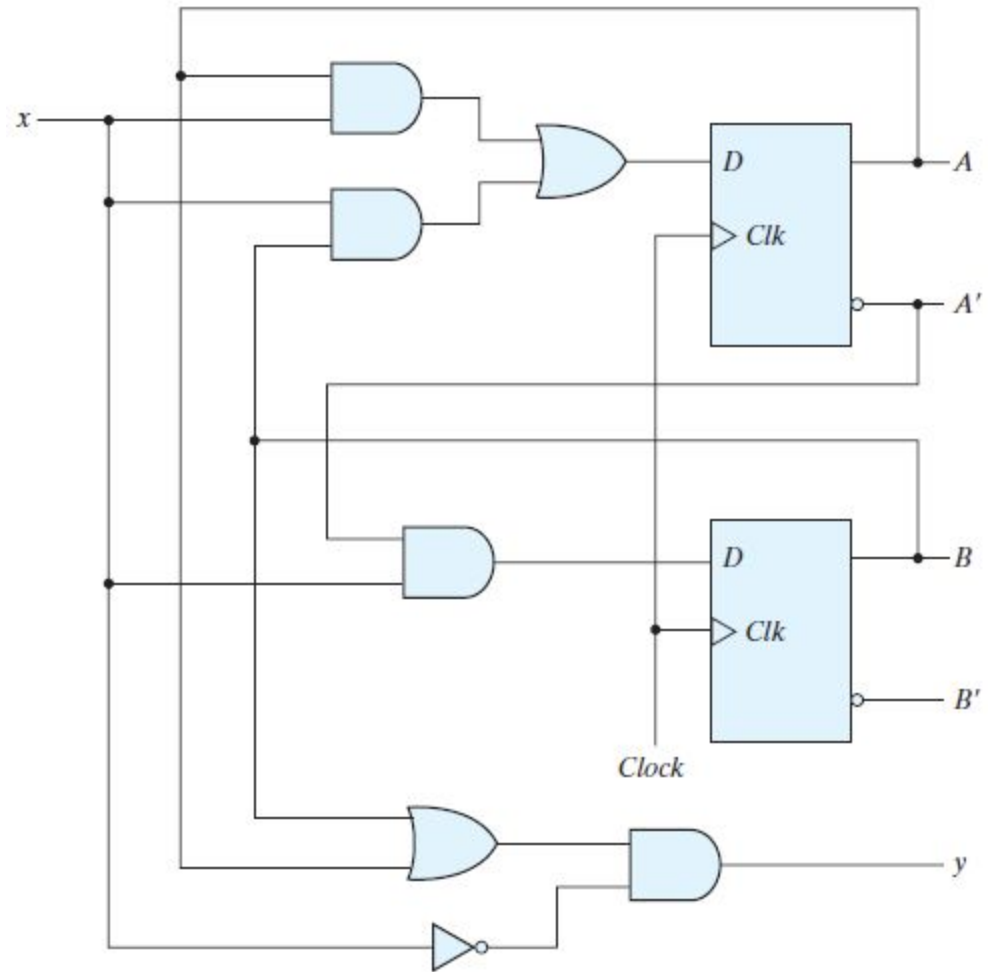
$$y(t) = (A + B)x'$$



Example 1 (Solution)

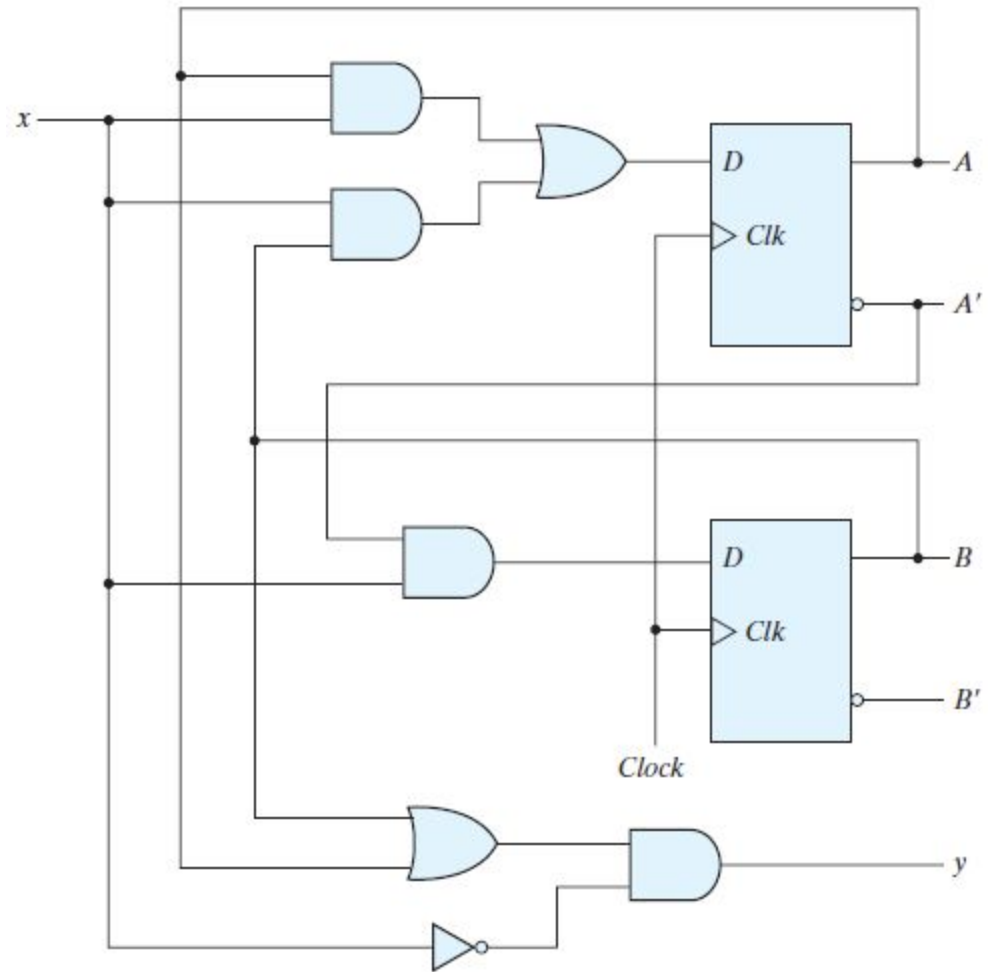
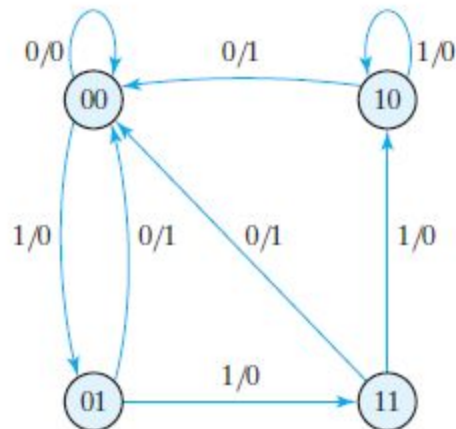
- State Table

Present State		Input x	Next State		Output y
A	B		A	B	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0



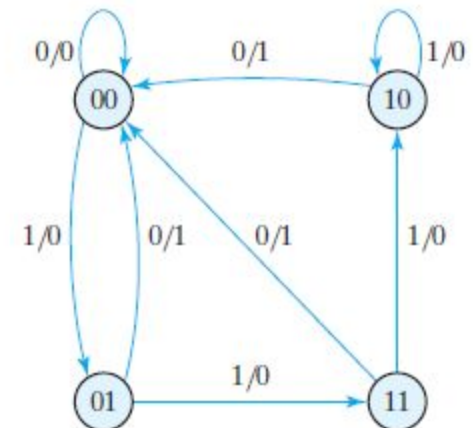
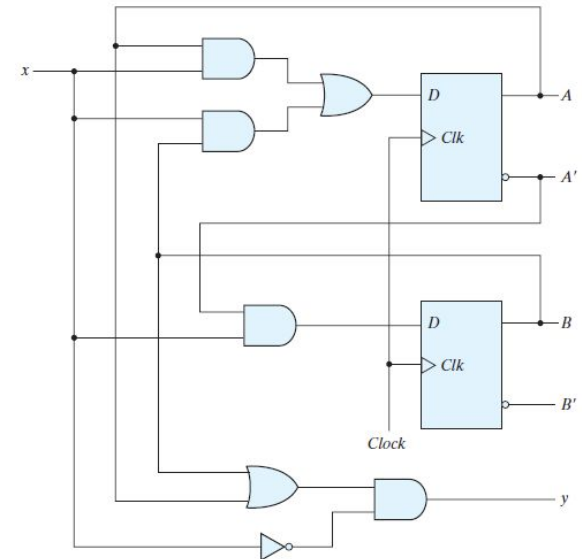
- State Diagram

Present State		Next State				Output	
		$x = 0$		$x = 1$		$x = 0$	$x = 1$
A	B	A	B	A	B	y	y
0	0	0	0	0	1	0	0
0	1	0	0	1	1	1	0
1	0	0	0	1	0	1	0
1	1	0	0	1	0	1	0



Example 1 (Solution)

- The state diagram clearly shows that, starting from state 00, the output is 0 as long as the input stays at 1
- The first 0 input after a string of 1's gives an output of 1 and transfers the circuit back to the initial state, 00
- **The machine represented by this state diagram acts to detect a zero in the bit stream of data**



Flip Flop Input Equations

- Sometimes instead of a circuit diagram a set of flip flop equations can be given
- From these flip flop equations we can easily construct the circuit and state table
- There are three types of analysis depending on flip flop equations
 - Analysis with D Flip Flop
 - Analysis with JK Flip Flop
 - Analysis with T Flip Flop

Analysis with D Flip Flop (1/2)

- Let us suppose that a circuit with the following D flip flop input equation is given

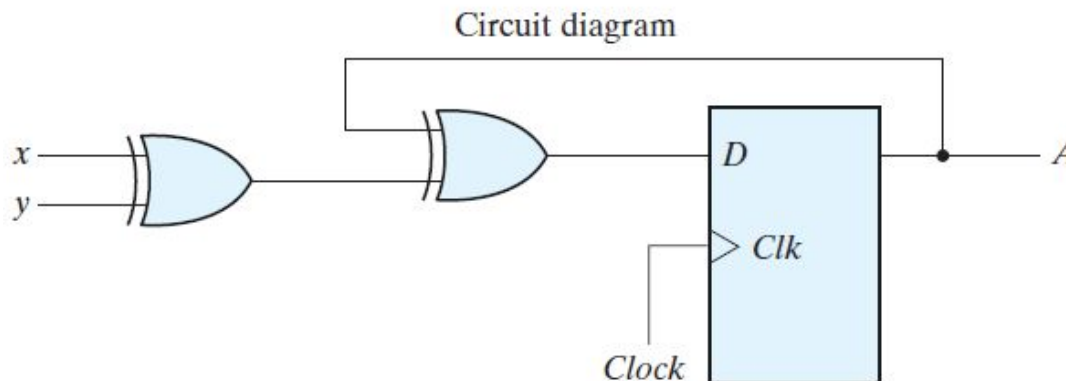
$$D_A = A \oplus x \oplus y$$

- Here:
 - D_A implies that it is a D flip flop
 - A is the present state
 - x and y are inputs
 - No equation for output is given, so the output of flip flop is the circuit's output

Analysis with D Flip Flop (2/2)

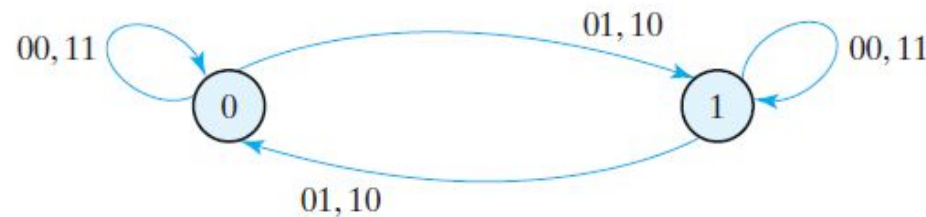
- For D flip flop $D = \text{next state}$, so

$$A(t + 1) = A \oplus x \oplus y$$



Present state	Inputs		Next state
A	x	y	A
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

State table



State diagram

Analysis with JK Flip Flop (1/3)

- Let us suppose that a circuit with the following JK flip flop input equations is given

$$J_A = B \quad K_A = Bx'$$

$$J_B = x' \quad K_B = A'x + Ax' = A \oplus x$$

- Here:
 - A and B shows that there are two JK flip flops
 - A and B are present states of flip flops
 - x is input
 - No equation for output is given, so the outputs of flip flops is the circuit's output

Analysis with JK Flip Flop (2/3)

- For JK flip flop the characteristic equations are given as

$$A(t + 1) = JA' + K'A$$

$$B(t + 1) = JB' + K'B$$

$J_A = B$	$K_A = Bx'$
$J_B = x'$	$K_B = A'x + Ax' = A \oplus x$

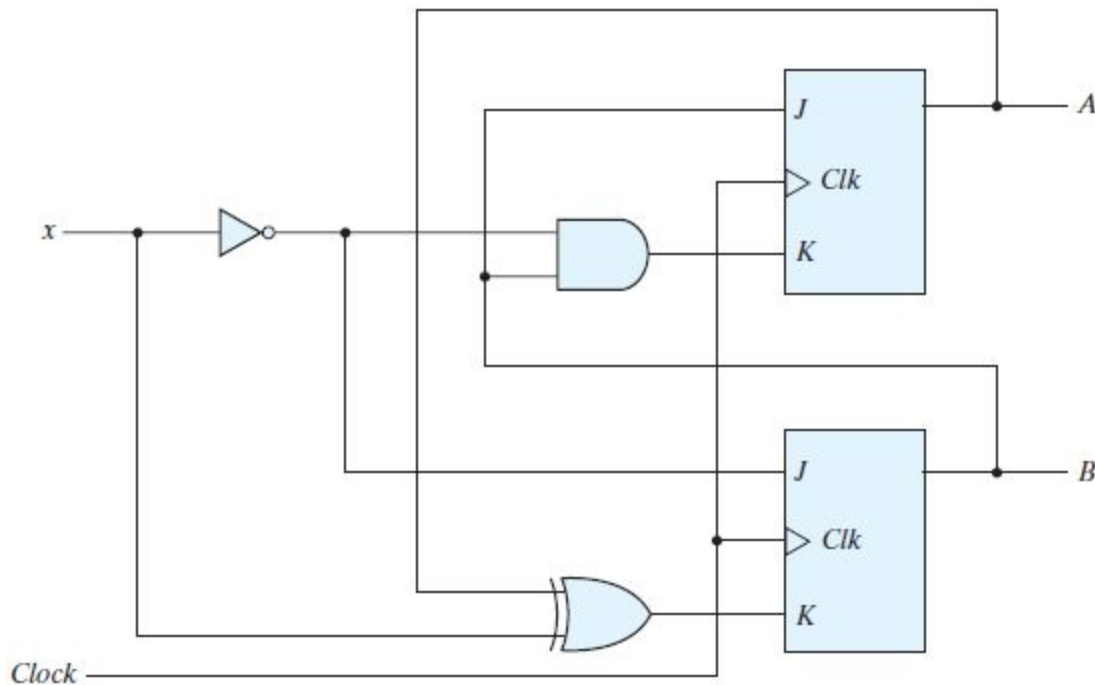
- Substitute the values of J_A , K_A , J_B , and K_B in above equations

$$A(t + 1) = BA' + (Bx')'A = A'B + AB' + Ax$$

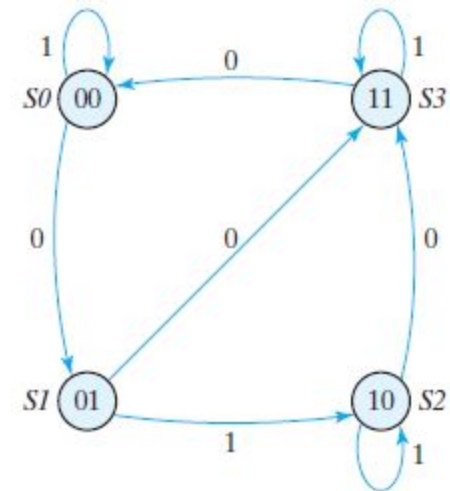
$$B(t + 1) = x'B' + (A \oplus x)'B = B'x' + ABx + A'Bx'$$

- The above two equations are state equations

Analysis with JK Flip Flop (3/3)



Present State		Input x	Next State	
A	B		A	B
0	0	0	0	1
0	0	1	0	0
0	1	0	1	1
0	1	1	1	0
1	0	0	1	1
1	0	1	1	0
1	1	0	0	0
1	1	1	1	1



Analysis with T Flip Flop (1/3)

- The analysis with T flip flop is same as the analysis with JK flip flop
- Let us suppose that a circuit with the following T flip flop input equations is given

$$T_A = Bx \quad T_B = x \quad y = AB$$

Here:

- A and B shows that there are two T flip flops
- A and B are present states of flip flops
- x is input
- y is output

Analysis with T Flip Flop (2/3)

- For JK flip flop the characteristic equations are given as

$$A(t + 1) = T'A + TA' = (T \oplus A)$$

$T_A = Bx$	$T_B = x$
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$$B(t + 1) = T'B + TB' = (T \oplus B)$$

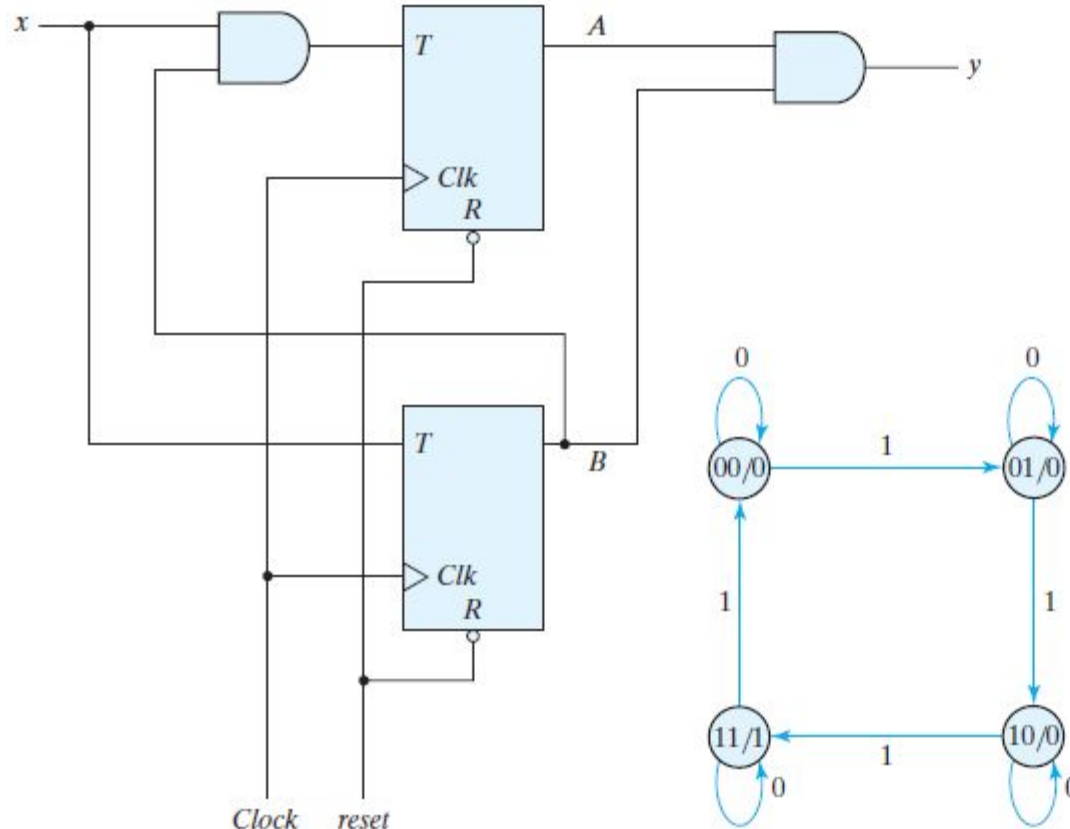
- Substitute the values of T_A and T_B , we get

$$A(t + 1) = (Bx)'A + (Bx)A' = AB' + Ax' + A'Bx$$

$$B(t + 1) = x \oplus B$$

- The above two equations are state equations

Analysis with T Flip Flop (3/3)



Present State		Input x	Next State		Output y
A	B		A	B	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	1	0
0	1	1	1	0	0
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	1